

# NETVISOR Mini Discovery V2

## Project Information

**Project Name:** NETVISOR

**Module:** Discovery Engine

**Version:** V2

**Author:** Moad

**Date:** 2025

---

## 1. Overview

NETVISOR Mini Discovery V2 extends the capabilities of V1 by introducing Layer 2 discovery mechanisms through ARP table analysis.

While V1 relied solely on ICMP Ping responses to identify active hosts, V2 enriches the discovery process by associating IP addresses with MAC addresses, providing a more accurate view of network assets.

This version represents the first transition from simple host discovery to actual asset identification.

---

## 2. Objectives

The objectives of V2 are:

- Maintain Ping-based host discovery
- Retrieve ARP cache information
- Associate IP addresses with MAC addresses
- Improve device visibility
- Prepare for future asset inventory capabilities

- Keep a modular architecture for future expansion
- 

## 3. Architecture

### Directory Structure

netvisor-mini-discovery-v2/

core/

- Network.php
- Scanner.php

modules/

- Ping.php
- Arp.php

services/

- DeviceBuilder.php

models/

- Device.php

run.php

---

## 4. Components

### 4.1 Device Model

Represents a discovered network asset.

Attributes:

- IP Address
- MAC Address

- Status
- 

## 4.2 Network Module

Responsibilities:

- Detect local IP address
- Determine network range
- Generate target IP list

Functions:

- `getLocalNetwork()`
  - `generateIPs()`
- 

## 4.3 Ping Module

Responsibilities:

- Send ICMP Echo Requests
- Identify active hosts

Functions:

- `pingHost()`
- 

## 4.4 ARP Module

New component introduced in V2.

Responsibilities:

- Read system ARP cache
- Extract IP-to-MAC mappings
- Provide Layer 2 information

Functions:

- `getARPTable()`

Data Source:

Linux Neighbor Table

Command used:

ip neigh

---

## 4.5 Scanner Core

Responsibilities:

- Execute Ping Sweep
- Retrieve ARP information
- Combine discovery results

Workflow:

Ping Sweep → ARP Collection → Device Creation

---

## 4.6 Device Builder

Responsibilities:

Transform raw discovery data into structured Device objects.

---

# 5. Discovery Workflow

## Phase 1: Network Detection

Determine local network.

Example:

192.168.1.0/24

---

## Phase 2: IP Generation

Generate all candidate hosts.

Example:

192.168.1.1 → 192.168.1.254

---

## Phase 3: Ping Sweep

Send ICMP requests to all targets.

Active devices are recorded.

---

## Phase 4: ARP Collection

Read ARP cache:

ip neigh

Example:

192.168.1.1 lladdr 64:29:43:70:68:16

---

## Phase 5: Correlation

Match:

IP → MAC

Example:

192.168.1.1 → 64:29:43:70:68:16

---

## Phase 6: Output

Display discovered assets with IP and MAC information.

---

## 6. Features Implemented

Implemented in V2:

- ✓ Single Network Discovery
  - ✓ ICMP Ping Sweep
  - ✓ Active Host Detection
  - ✓ ARP Cache Analysis
  - ✓ MAC Address Collection
  - ✓ IP-MAC Correlation
  - ✓ Modular Architecture
  - ✓ CLI Execution
  - ✓ Layer 2 Visibility
- 

## 7. Example Output

Example:

IP: 192.168.1.35  
MAC: 7c:2a:db:ee:3a:03

IP: 192.168.1.1  
MAC: 64:29:43:70:68:16

IP: 192.168.1.38  
MAC: 0e:24:2c:a3:3e:23

Total Devices: 4

---

## 8. Improvements Over V1

| Feature               | V1      | V2       |
|-----------------------|---------|----------|
| Ping Discovery        | ✓       | ✓        |
| Active Host Detection | ✓       | ✓        |
| MAC Address Detection | ✗       | ✓        |
| ARP Analysis          | ✗       | ✓        |
| Asset Identification  | Limited | Improved |
| Layer 2 Visibility    | ✗       | ✓        |

---

## 9. Current Limitations

Not yet implemented:

- ✗ Hostname Resolution
  - ✗ Vendor Detection (OUI Lookup)
  - ✗ TCP Port Scanning
  - ✗ Service Detection
  - ✗ Operating System Fingerprinting
  - ✗ Device Classification
  - ✗ Database Storage
  - ✗ Multi-Network Discovery
  - ✗ Topology Mapping
  - ✗ Monitoring Engine
-

# 10. Technical Challenges Encountered

During development, several issues were identified:

- Empty ARP cache results
- Incorrect parsing of ARP entries
- Missing IP extraction from neighbor table
- Docker bridge interfaces appearing in results
- Inconsistent device visibility depending on network traffic

These issues were progressively resolved through debugging and parser improvements.

---

```
NETVISOR V2 - REAL DEVICE DISCOVERY

Phase 1: Ping Sweep...
Phase 2: ARP Table...
===== DEVICES FOUND =====

IP: 192.168.1.2 | MAC: unknown
IP: 192.168.1.42 | MAC: unknown
IP: 192.168.1.56 | MAC: unknown
IP: 192.168.1.80 | MAC: unknown
IP: 192.168.1.102 | MAC: unknown
IP: 192.168.1.105 | MAC: 0e:24:2c:a3:3e:23
IP: 192.168.1.107 | MAC: unknown
IP: 192.168.1.110 | MAC: unknown
IP: 192.168.1.113 | MAC: unknown
IP: 192.168.1.116 | MAC: unknown
IP: 192.168.1.120 | MAC: unknown
IP: 192.168.1.122 | MAC: unknown
IP: 192.168.1.123 | MAC: unknown
IP: 192.168.1.128 | MAC: unknown
IP: 192.168.1.129 | MAC: unknown
IP: 192.168.1.156 | MAC: unknown
IP: 192.168.1.157 | MAC: unknown
IP: 192.168.1.254 | MAC: e0:3d:a6:95:4b:50

TOTAL: 18
```



# 11. Future Roadmap

## Version 3

- TCP Discovery
- Open Port Detection
- Basic Service Enumeration

## Version 4

- Hostname Resolution
- Vendor Identification

## Version 5

- Multi-Network Discovery

## Version 6

- Asset Inventory Database

## Version 7

- Topology Engine

## Version 8

- Monitoring Engine

## Version 9

- Web Dashboard

---

# 12. Conclusion

NETVISOR Mini Discovery V2 successfully expands the discovery engine beyond simple host detection by introducing Layer 2 asset identification through ARP analysis.

This version establishes the foundation for future asset inventory and network topology capabilities while maintaining a clean and extensible architecture suitable for enterprise-scale development.